Practical exercise 3

1. Consider again the Stamford `ozone` data. Use `R` to perform a chi-square goodness-of-fit test to decide whether the ozone concentration observations in Stamford may come from a normal distribution.

   Hint: The answer is given in the lectures – follow the procedure in the notes, and see `R` code used in lectures on Canvas. In `R`, first use the `cut(...)` and `table(...)` commands to obtain the observed frequencies in the 10 considered cells. Then the `pnorm(...)` command can be used to calculate expected frequencies in the cells. Compute the chi-square statistic and then the p-value using the `pchisq(...)` command.

2. More on the χ^2 goodness-of-fit test.

   The χ^2 goodness-of-fit (gof) test described in lectures can also be performed by splitting the range of the tested distribution into k intervals (cells) of **equal probability** (rather than equal length). For example, if we want 4 equal-probability intervals for the $N(0,1)$ distribution, these are given as

   $$(-\infty, -0.6745), (-0.6745, 0), (0, 0.6745), (0.6745, \infty).$$

   You can check that the probability of the $N(0,1)$ distribution in each of these intervals is 0.25. You can obtain the break points of the intervals from statistical tables, or in R by using

   `qnorm(0), qnorm(1/4)`, etc.

   The test then proceeds as in the equal-length cells case.

   (a) For the `ozone` data, use the χ^2 test with 10 *equal-probability* intervals to decide whether the ozone concentration observations in Stamford come from a normal distribution. Use the file `"R code for norm.chi2.gof.R"` given on canvas to get the results. You must understand R codes used in `norm.chi2.gof( )` function.

   Remark: It would be useful to write a function in `R` that performs the test you wish to check. You can modify R codes given in `norm.chi2.gof( )` function to check the χ^2 goodness of fit test of other distribution.

   (b) You will notice that in part (a) the test gives p-value $= 1.9\times10^{-5}$, which is different than the p-value obtained when the test was performed with 10 equal-length cells. Is the result obtained in Q1 good to use? Explain.

   Repeat the test using a different number of cells or different break points (in the case of equal-length cells) and notice that the p-value is different every time.

   Remark: This highlights a disadvantage of the χ^2 gof test. It is *sensitive* to the choice of intervals. In order to minimise the arbitrariness in choosing these intervals, it is preferable to use equal-probability cells to perform the test (as this only requires choosing the number of cells and not the break points.)

(c) Next, install `nortest` package in `R`, then use `pearson.test( )` given in `nortest` R library to check the χ^2 goodness of fit test of Stamford data. Comment on your results. You can run R syntax line by line as follows:

```
> install.packages("nortest")
> library(nortest)
> pearson.test(data, n.classes)
```

3. χ^2 goodness-of-fit for other distributions.

The data in file `call_centre_data.txt` are a sample of times (in minutes) between 3 phone calls received at a company's customer services call centre at busy periods of the day.

(a) Draw a QQ plot to explore the fit of a Gamma(3,2) distribution to the data.

(b) A statistician suggests a Gamma(3,1) distribution instead. Justify this option, based on the QQ plot from part (a).

(i) Perform a χ^2 gof test for the Gamma(3,1) distribution. You should initially use 7 intervals of length equal to 1.5 for the test (starting from 0), and then combine cells if required.

(ii) Next, conduct a χ^2 goodness-of-fit test for the Gamma(3,1) distribution using 7 intervals of equal probability. Compare your result with the answer from 3(b)(i).

4. Kolmogorov-Smirnov g-o-f test

The `R` code below performs a K-S test for the hypothesis that data follow a N(0,1) distribution. Use the code in `R` to understand how the test works (also see lecture notes).

```
============================================================
## K-S test for N(0,1)
============================================================
help(ks.test)

## Generate sample from N(0,1)
n = 15
y = rnorm(n,0,1)
par(mfrow=c(2,1))
stripchart(y,pch=19);
boxplot(y, horizontal=TRUE);

##  Perform KS test
ks <- ks.test(y,"pnorm",mean=0,sd=1); ks
## Extract statistic
D <- ks$statistic; D
```

```
## Extract p-value
pval <- ks$p.value; pval
##    Plot ecdf and cdf
grid.x = seq(min(y),max(y), length=100)
plot(grid.x,pnorm(grid.x),type="l",col="red", ylim=c(0,1))
s = c(1:n)
lines(sort(y),s/n, type="s")
title(main="ECDF and N(0,1) CDF")

## Try a sample from other distribution
## eg from t(1)
y2 = rt(n,1)
stripchart(y2,pch=19);
boxplot(y2, horizontal=TRUE);
ks <- ks.test(y2,"pnorm",mean=0,sd=1); ks

## or from t(5)
y3 = rt(n,5)
stripchart(y3,pch=19);
boxplot(y3, horizontal=TRUE);
ks <- ks.test(y3,"pnorm",mean=0,sd=1); ks
```

5. Consider the data on determination of copper in wholemeal flour (data set `chem` within the MASS library). Use the K-S test to investigate the hypothesis that the data come from an exponential distribution with mean 3.68μg g^{-1}.